Avionics Cooling Fan – General data

1. General

To provide cool air circulation in the forward avionics compartment, located in the nose between stations 37.21 to 44.00. Cool air is drawn into the compartment through an intake and circulated around the compartment by two electronic fans installed on the top avionics shelf.

2. Description

Two Sandia SAFE 128 avionics cooling fans are installed in the forward avionics compartment to provide air circulation for the avionics. The fans include a fault detection output which provides a ground when the fans are operating above a preset RPM threshold. Two relays, located inside the junction box, are triggered by the ground signal from each fan output. If either relay opens, a ground is provided to the MAU, which produces an Avionics Fan Fail CAS.

The avionics cooling fans are mounted on the upper avionics shelf around station 54, one on the pilot's side, another on the co-pilot's side.

28 Volt DC power is provided from the left bus through a 1 amp AVIONICS FAN (E7) circuit breaker on avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

3. Adjustment/Test

A. Avionics Cooling Fan

The unit may be tested by manually slowing the fans, one at a time, by applying a light pressure to the center hub of the fan until the Avionics Fan Fail CAS appears. Confirm the CAS clears when the fan returns to full speed.

4. Removal/Installation

A. Removal

1) Ensure the aircraft electrical system is OFF.

2) Pull circuit breaker AVIONICS FAN (E7) (left bus) on the avionics circuit breaker panel and fit safety clip.

3) Gain access to the forward avionic shelves through the forward baggage compartment.

4) Remove top and middle covers attached to bulkhead station 37.21 and 44.00 respectively for full access.

5) Disconnect the electrical connections.

6) Remove four mounting screws.

7) Remove fan.

Note Stow connectors of removed unit.

B. Installation

1) Ensure the aircraft electrical system is OFF.

2) Install fan on the upper avionics shelf.

3) Set unit in position and install four mounting screws.

4) Connect the electrical connectors. Ensure connection is sound and secure.

5) Reinstall top and middle covers attached to bulkhead station 37.21 and 44.00 respectively. Ensure covers are mounted correctly and secure.

6) Remove safety clips and reset circuit breaker AVIONICS FAN (E7) (Left Bus) on the avionics circuit breaker panel.

7) Close forward baggage compartment door.

5. Troubleshooting

A. Avionics Fan Fault

1) Ensure AVIONICS FAN (E7) circuit breaker is closed.

2) Visually inspect the connector for damaged wiring or contacts. Ensure electrical connector is fully seated.

3) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

4) If repair or service is required, send to appropriately rated avionics repair facility.